3.5inch DPI LCD

Introduction

640x480, 3.5inch IPS Display for Raspberry Pi, DPI interface, 5-points capacitive touch control.

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Features

- 3.5inch IPS Display , 640x480 hardware resolution.
- Designed for Raspberry Pi, based on Raspberry Pi 40pin GPIO interface.
- Using DT6566 interface of Raspberry Pi to drive LCD, refresh rate up to 60Hz.
- Drive capacitive touch through I2C interface of Raspberry Pi, 5-points touch, toughened glass panel, hardness up to 6H.
- Supports Raspbian and Kali systems.

Software settings

If you don't know the difference between the two branches of Raspberry Pi OS, you can check the following introduction #Introducing the Raspberry Pi OS fork

For Raspberry Pi OS Bullseye branch

Please download the latest version of the image from the Raspberry Pi official website.

1. Download the compressed file to the PC, and extract the img file.
2. Connect the TF card to the PC and use SDFormatter.exe to format the TF card.
3. Open the Win32DiskImager software, select the system image prepared in step 1, and click write to burn the system image.
4. After the programming is completed, open the config.txt file in the root directory of the TF card, add the following code at the end of config.txt and save it.

```
dtoverlay=vc4-kms-v3d
dtoverlay=vc4-kms-DPI-ili9486
dtoverlay=waveshare:70pin-35
dtoverlay=waveshare:ili9481-35
dtoverlay=waveshare:70ds-35
display_default_lcd=1
extra_transpose_buffer=2
dpi_group=2
dpi_mode=87
dpi_output_format=0x07f216
hdmi_timings=640 0 20 10 10 480 0 10 5 5 0 0 0 60 0 40000000 1
dtoverlay=waveshare:70dpi-35-b
dtoverlay=waveshare:70dpi-35
dtoverlay=waveshare:70ds-35
```

5. Download the 3.5inch DPI LCD DTBO file and extract the dtbo files. Copy these files to the overlays directory (/boot/overlays/).
6. Insert the TF card into the Raspberry Pi, power on the Raspberry Pi, and wait for more seconds.
7. After the programming is completed, open the config.txt file in the root directory of the TF card, add the following code at the end of config.txt and save it.

```
dtoverlay=vc4-kms-v3d
gpio=2-17=a2
gpio=20-27=a2
dtoverlay=ugt10
disable_mc_ir=0
display_default_lcd=1
extra_transpose_buffer=2
dpi_group=2
dpi_mode=87
dpi_output_format=0x07f216
hdmi_timings=640 0 20 10 10 480 0 10 5 5 0 0 0 60 0 40000000 1
dtoverlay=waveshare:70dpi-35-b
dtoverlay=waveshare:70dpi-35
dtoverlay=waveshare:70ds-35
```

8. Download the 3.5inch DPI LCD DTBO file and extract the dtbo files. Copy these files to the overlays directory (/boot/overlays/).
9. Save and quit the TF card safely, and insert the TF card into the Raspberry Pi.
10. Insert the 3.5inch DPI LCD into the 40PIN GPIO interface of the Raspberry Pi, power on the Raspberry Pi, and wait for about ten seconds to display normally.

Rotation(Working with Raspberry Pi)

Raspberry Pi OS and Kali display rotation

First check whether the KMS or FKMS driver is loaded on the system you are using.

Check method: In /boot/config.txt, check whether the line dtoverlay=vc4-kms-v3d or dtoverlay=vc4-fkms-v3d is turned on.

With KMS or FKMS driver loaded

Use the following command for display rotation:

```
sudo nano /etc/xdg/lxsession/LXDE-pi/autostart
#Modify the command corresponding to the display rotation angle at the end of
the autostart file, and it will take effect after rebooting the system.
#1: rotate 90 degrees; 2: rotate 270 degrees; 3: rotate 180 degrees; 0: rotate
0 degrees
kcalib -p 1
```

No KMS or FKMS driver loaded

Use the following command for display rotation:

```
sudo nano /boot/config.txt
#Use the command corresponding to the display rotation angle at the end of
the config.txt file, and it will take effect after rebooting the system.
#1: rotate 90 degrees; 2: rotate 270 degrees; 3: rotate 180 degrees; 0: rotate
0 degrees
display_rotate=1
```

Touch Rotating

After the display is rotated, the position of touch is incorrect because the touch doesn't change with the display angle. So the touch also needs to be modified.

1.Install libinput.

```
sudo apt-get install xserver-xorg-input-libinput
```

If the system you installed is Ubuntu or Jetson Nano. The installation code is:

```
sudo apt install xserver-xorg-input-evdev
```

2.Create the xorg.conf.d directory under /etc/X11/ if the directory already exists, proceed directly to step 3).

```
sudo mkdir /etc/X11/xorg.conf.d
```

3.Copy the 40-libinput.conf file to the directory you created just now.

```
sudo cp /usr/share/X11/xorg.conf.d/40-libinput.conf /etc/X11/xorg.conf.d/
```

4.Edit this file.

```
sudo nano /etc/X11/xorg.conf.d/40-libinput.conf
```

Find the part of the touchscreen, add the following statement inside, and then save the file.

```
Option "CalibrationMatrix" "0 1 0 -1 0 1 0 0 1"
```

Similar to the picture below:

5. save and reboot your Pi

```
sudo reboot
```

After completing these steps. The LCD could rotate 90 degrees in both display and touch function.

Note:

90 degree: Option "CalibrationMatrix" "0 1 0 -1 0 1 0 0 1"

180 degree: Option "CalibrationMatrix" "-1 0 1 0 -1 1 0 0 1"

270 degree: Option "CalibrationMatrix" "0 -1 1 1 0 0 0 0 1"

Others

Disable power saving

If you want to keep the display turning on all the time, you can disable the power saving function.

Modify file lightdm.conf

```
sudo nano /etc/lightdm/lightdm.conf
```

Find the [SeatDefaults] option and uncomment the line"xserver-command", modify it as follows:

```
#xserver-command=X
```

Modify this code to

```
xserver-command=X -s 0 -dpms
```

- -s 0 –Disable the display protecting.
- dpms Disable power saving.

Reboot

```
sudo reboot
```

Control brightness by PWM

If you use PM, you need to update wiringpi first:

```
wget https://project-downloads.drogon.net/wiringpi-latest.deb
sudo dpkg -i wiringpi-latest.deb
gpio -v
# Check if it is 2.52 version
```

Control the brightness by PWM

```
gpio -g mode 18 pwm        #Configure the PIN as PWM mode
gpio pwm 1\12              
gpio -g pwm 18 1           #Brightest
gpio -g pwm 18 1023        #Slowest
gpio -g mode 18 out        #Clear the PIN to output mode.
```

Turn on/off HDMI output

You can use the following commands to disable/enable the displaying.

```
vcgencmd display_power 0
vcgencmd display_power 1
```

Introducing the Raspberry Pi OS fork

On December 2, 2021, the Raspberry Pi OS for Raspberry Pi split into two branches, the Buster branch and the Bullseye branch.

The Buster branch is a continuation of the old system, which is more stable and uses raspicam as the default camera subsystem.

The Bullseye branch adds some new features, including KMS drive as the default display driver and libcamera as the default camera subsystem.

Image download address

Operating system images

Many operating systems are available for Raspberry Pi including Raspberry Pi OS, our official supported operating system, and operating systems from other organisations.

Raspberry Pi Imager is the quick and easy way to install an operating system to your Raspberry Pi. Imager includes a wide range of operating systems, including many third-party operating systems. Alternatively, choose from the operating systems below, available to download and install manually.

Download
Raspberry Pi OS
Raspberry Pi OS (32-bit)
Raspberry Pi OS (64-bit)
Raspberry Pi OS Lite
Third-party software

More detailed information can be found at:

Raspberry Pi Branch Details

Buster Release Notes

Bullseye Release Notes

Notes:The above information was updated on January 24, 2022. Due to the frequent update of the Raspberry Pi system, we can't see the this information, the relevant information may be out of date. The actual performance of the image shall prevail. This is for reference only.

Interface

The pins labeled "NC" below means that those pins of Raspberry Pi are not occupied by the LCD, and the user can use them for other applications.

PIN NO.	SYMBOL	DESCRIPTION
1	NC	Power positive (3.3V power input)
2	5V	Power positive (5V power input)
3	VSYNC	Vertical sync
4	5V	Power positive (5V power input)
5	HSYNC	Horizontal sync
6	GND	Ground
7	B2	RGB signal cable
8	G4	RGB signal cable
9	GND	Ground
10	G5	RGB signal cable
11	G7	RGB signal cable
12	PWM	Backlight control
13	TP_INT	Touch Panel interrupt
14	GND	Ground
15	R4	RGB signal cable
16	R5	RGB signal cable
17	NC	Power positive (3.3V power input)
18	R6	RGB signal cable
19	TP_SDA	I2C SDA for touch panel
20	GND	Ground
21	B7	RGB signal cable
22	R7	RGB signal cable
23	TP_SCL	I2C SCL for touch panel
24	B6	RGB signal cable
25	GND	Ground
26	B5	RGB signal cable
27	LCD_CLK	LCD clock
28	DE	LCD data enable
29	B3	RGB signal cable
30	GND	Ground
31	B4	RGB signal cable
32	G2	RGB signal cable
33	G3	RGB signal cable
34	NC	GPIO interface
35	NC	GPIO interface
36	G6	RGB signal cable
37	NC	GPIO interface
38	R2	RGB signal cable
39	GND	Ground
40	R3	RGB signal cable

Resource

3D Drawing

- 3.5inch DPI LCD 3D Drawing

Software

- Panasonic SDFormatter
- Win32DiskImager
- PuTTY

FAQ

Anti-Piracy

Since the first-generation Raspberry Pi released, Waveshare has been working on designing, developing, and producing various fantastic touch LCDs for the Pi. Unfortunately, there are quite a few pirated/knock-off products in the market. They're usually some poor copies of our early hardware revisions, and comes with none support service.

To avoid becoming a victim of pirated products, please pay attention to the following features when purchasing:

(Click to enlarge)

Beware of knock-offs

Please note that we've found some poor copies of this item in the market. They are usually made of inferior materials and shipped without any testing.

You might be wondering if the one you purchased is genuine or you've purchased in other non-official stores is original, feel free to contact us.